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HARRIER JET JUMP FLAP AND NOSECONE FORMULA 1 CAR MOULD DESIGNS FOR A COMPOSITE DEVELLOPEMENT

-End of Study Project (B.Eng) -

Steeve SAAH

Under the care of

M.GLENN MILES

END-OF-STUDY PROJECT

DOCUMENT'S TITLE :	Harrier Jet Jump Flap and Nosecone Formula 1 Car Mould Design For a Composite Development
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ABSTRACT:

The Nosecone has permitted important technological avances such purpose with the conquest of the Moon. Since aircraft Dassault France has always claimed its military independence .In this context the need to increase the reach of missiles is necessary. The Nosecone is first technology component in contact in compressible middle is the nosecone whose performances are optimal to come into atmospheres.

Transport industries are those that automotive and aerospace companies have advocate the study of the Nosecone shape from a composite form to improve their products.

In this context, Automotive and Aerospace engineers of the next generation must be familiarized with the Nosecone Technology.

This report presents Research undertaken at Coventry University to develop a composite version of the Nosecone and the Flap. The Design Engineering tools are essentials to generate accurate dimensions.

Resources offered by the Dassault software CATIA V5 are adapted to accomplish our tasks.

Keywords: Nosecone, Flap, Design

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ACKNOWLEDGEMENTS

I would like to thank Mr Miles Glenn Miles for encouraging and supporting me during this interesting project. I felt honored for being associated to such prolific project such the Design and Composite Development for the Automotive and Aeropace department of Coventry University.

LITTERATURE REVIEW

Books

- CATIA Core Tools-Computer Aided Three Dimensional Design written by Michel Michaud
- Chevalier Design Industrial Guide
- GUHRING milling tools

Websites

1. https://www.youtube.com/watch?v=Fg2EstV4HgU

- https://www.youtube.com/watch?v=G9wv6g7Gx20
- https://www.youtube.com/watch?v=oRElMGa29EU
- https://www.youtube.com/watch?v=VKtw214RFZI
- http://catiadoc.free.fr/online/CATIAfr_C2/dseugCATIAfrs.htm
- https://www.youtube.com/watch?v=VodfQcrXpxc

CONTEXT

Who are the product's users?

The products users are the Composite material Professor Mr Miles Glenn but also the research staff member from an industry can also be interested by the different mould (mould composite and Aluminium mould).

A company is interested by the seepage gas. This company can buy our data (Latex curing time during the experience). This company s interested in about how we can increase their pipe pressure to improve the quality. Once the pressure has been increase how we can maintain this one constant.

What are the prospects?

The product has the perspective to move towards an autoclave. It will allow the understanding of the process by autoclave by students. The autoclave process allows the tradeoff between the price it generates for its use and understanding of the process. The university does not need to generate expenditure in important educational aim.

Vacuum molding techniques will improve the quality of the latex for the manufacture of the mold. Latex objects with the most latex complex shapes can be created as well .In the near future, the project will allow us to build a new mold with different internal characteristics (large elastic resistance ...). We can now manufacture products generating forms through high temperature and pressure through lower cost.

Those to which we are interested is the composite structure of the "Nosecone" a laboratory racing car developed from ever developed prior process. This is the occurrence of manufacturing epoxy based and latex.

What are the conditions of disposal of the product?

The product can be stored and used indefinitely.

Diagnostics market

what are the disadvantages of similar products?

The following product has never been manufactured .The disadvantage of similar products cannot be defined.

What are the competing products? What aspects need changing?

There are no competing products.

What is the estimated product cycle?

This is a research work .It is estimated that the university will use the product for many years. The data we collect is important.

What benefits do we expect in the Task book functional specifications?

There are two types of benefits. The mould can be used for teaching activities but the composite version developed can have industrial application in the automotive industry.

Specific directives Product

what are the design requirements?

The mold has to be composed of 4 blocks of female molds, and one block of male mould. The thickness between the male and the female mould, once assembled , has to be three times the thickness of the original nosecone designed by the Coventry motorsport Student.

What is the amount of predictable investments and over what period?

The amount of investment is 3 510 euros over a period of three months.

The time is three months from 30 May to 1 September 2016 to carry out study of the composite structure of the "nosecone" directed based epoxy and latex. The project can be continued for two more months.
The parameters that we consider are weight "nosecone" the Aluminum material properties, the inflation pressure of the female mold, temperature and curing time of the latex / epoxy.

What are the health and safety requirements?

When making the mold manufacturer shall be provided with a gown and safety boots.

What are the expectations in terms of ergonomics?

The space aluminum mold must fit perfectly into the surface of the "nosecone".
It provide a smooth wall of 3.6 mm, three times the thickness of the existing "nosecone" .
The metal mold must be divided into four blocks, each reusable according to the research needs.

INTRODUCTION

This project is incorporated within the framework of an internship agreement between Coventry University and the University Paris 13 for being awarded a Bachelor Degree of Mechanical engineering.

Thus, this paper describes my three months' work at the Computer Engineering Department of Coventry University as a Research Assistant to apply the knowledge on Design Engineering that I have earned during a one year study at the University Paris 13.This experience provided the opportunity to ally theoretical understanding and practical skill necessary to work in a Manufacturing Industry.
This report presents the tests that have been conducted as part of the design process, how the design engineering added value to the research, and to conclude the model Job's analysis is used to explain how Coventry University will take advantage of our results.

PRESENTATION

Our Research is going to take at Coventry University.

- University historic

1843: Creation of Coventry University with the College Coventry of Design.
1950: Coventry school of Art is renamed College of Art.
1961: To register more Student in at the City College Coventry (an Engineering training at Coventry University, Coventry University is renamed College Lanchester of Technology named to honor. Frederick Lanchester a polymath and automotive engineer.
1971: Coventry University is designated as institution by The Education Secretary Margaret Thatcher.

University of Coventry is an English Public Research University located in England. It is largest one of the city.

This university is a member of the Association of Commonwealth universities, the University Alliance and Universities UK. This institution has recently received several awards.
Indeed, during the years 2014,2015,and 2016, Coventry University has been named Modern University of the Year by the Times and Sunday Times University Guide and is particularly distinguished for its Engineering and Automotive Design Courses.
In 2015, Coventry University is ranked 15th among the best University of the United Kingdom.

The University is composed is of 23 Buildings. Some them are described below.

- The Allan Berry Building which is used for the registry and the business development
- The Alma Building which is used as home for the professional services.
- The Armstrong Siddeley which is used which for the design and the build of cars and Aircraft to develop an industrial and practical experiments.
- The Bugatti is used as area for Automotive and transport Design .This building has a very Strong relationship with industries.
- The Engineering and Computing Building which is used as research laboratory by student And Professors. This building is equipped of developed teaching technologies as the wind Sponsored by Mercedes Mc Laren and a Harrier Jet Jump Airplane.
- The Ellen Terry Buildings which used by the Faculty of Art and Humanities
- The Frederick Lanchester Library which used by the university members. This building Is equipped by computers and books to help out to facilitate the research tasks.
- Graham Sutherland
- The Hub is space to welcome visitors and student .This building also provides doctors and Surgery for the student, staff members and visitors.
- The Jaguar Building
- The Maurice Foss is mainly used by Automotive Design research. This area is also Equipped by tests benches and a manufacturing laboratories.
- The Multi-Storey Car Park which is a car park for Staff and visitors.

METHODS

In order design our mould, three nosecone have been considered.
The design properties if each model were studied to select the most appropriate one.
Methods.

The model selected, the nosecone car is presented in figure 1 below.

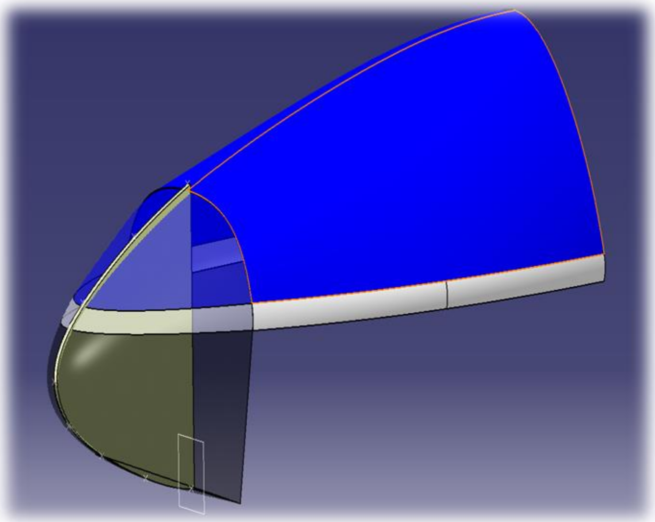


Figure-1.The 2015 Nosecone

The Nosecone Car

The nosecone Car is the front of the car designed to offer the minimum aerodynamic resistance by reducing the trailing efforts of the vehicle during its motion into a incompressible fluid. The shape in form of solid of revolution must be designed and kept so that we can keep optimum specifications. The friction produced by the movement of the nosecone in the air is similar to that produce by a bullet. Both of them are revolution solids. This why preliminary study is simulating an impact of bullet by using Abaqus.

- **Preliminary Study : The impact of bullet**

To simulate the impact of a bullet on a wall we create two geometrical shapes, about which we applied as aluminium materials. In order to simulate the high velocity damage we defined the Johnson cook parameters (Abaqus tutorial 9 2013) on the bullet that we consider the front of this as solid of revolution. These parameters are used to defined criteria from a bullet impact in high velocity damage.

Figure-1.1

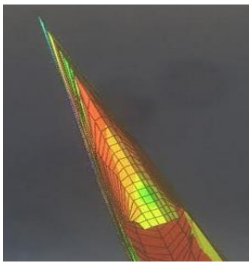


Figure-1.2

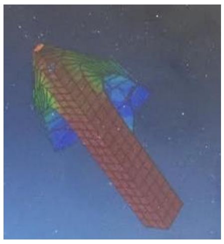


Figure-1.3

We can observe that as and when time runs, the wall progressively moulds the shape of bullet. This wall finally espouses the shape of the bullet. At the end of the simulation the moulded shape has been retained by the wall. Therefore on our mould design concern, the major challenge is to be able to design a mould that can be used to generate the original shape with the greatest accuracy possible.

CAD 2014 Model Investigation

This project was based on the nosecone CAD Model produced by 2014 Formula Student team. (See figure 1) using CATIA V5.

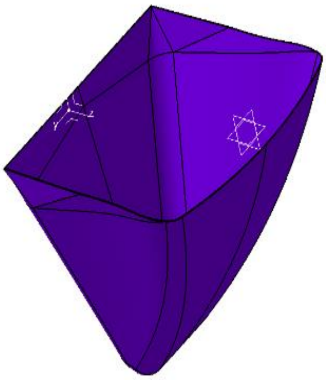


Figure-2

These are the dimensions of the CAD Model

Dimensions:

- Height:354.944 mm (See figure -2.1)
- Width: 382.9 mm   (See figure -2.2)
- Mould thickness: 1.6mm (See figure -2.3)

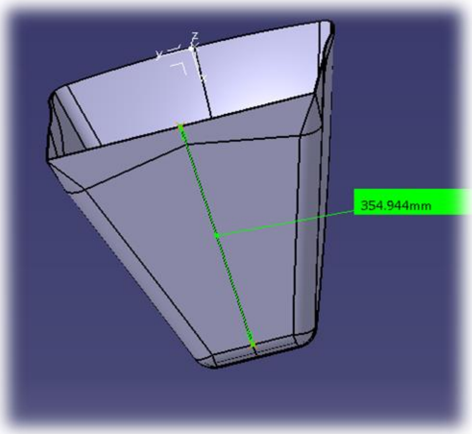


Figure-2.1

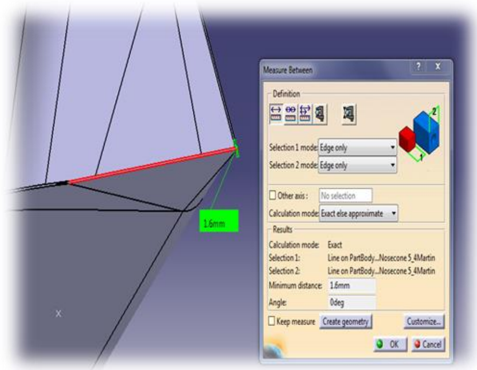


Figure -2.2

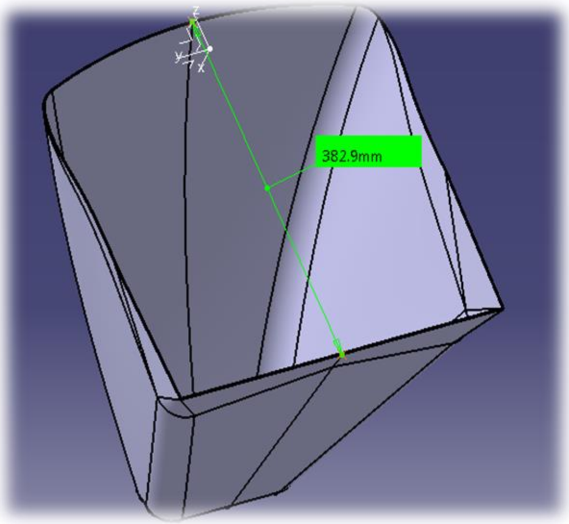


Figure-2.3

Construction problems

- The Nosecone 2014 design observations revealed several constructions irregularities that are presented in the figures below.

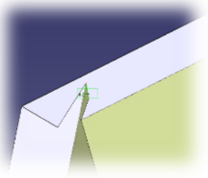


Figure-3.1- Surface Construction problem

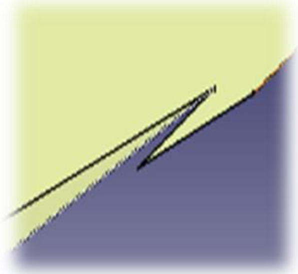


Figure-3.2- Surface Contour Construction Problem

After zooming we could find out many miscutted sections at the top of the Nosecone.

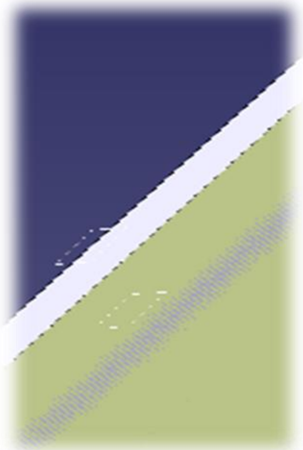


Figure-3.3 Cutting plan creation

These irregularities needed to be solved to avoid errors in the design of the female and male nosecone.

Solution

To solve the problem a one millimetre plan has been created from the upper surface of the nosecone extracted that we used to split the whole contour of this upper surface. This operation could be done before extraction, we would obtain the result in terms of constant and smooth contour. (See Figure 4.1, 4.2, 4.3, below)

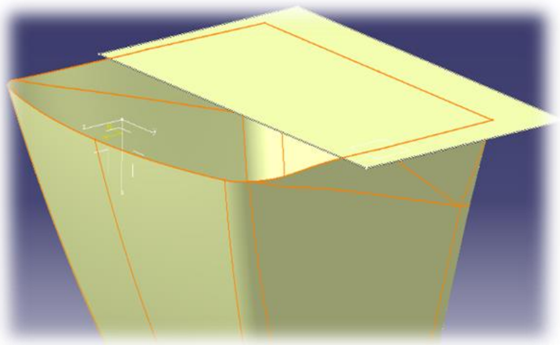


Figure-4.1-Cutting plan creation covering only the problematic surface

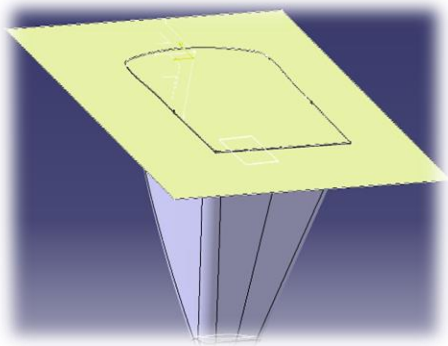


Figure-4.2-. the 1 mm cutting plan

Male Mould Design

To build up the male mould, the generative shape Design workshop was used
At first step the intern surface of the mould was extracted by using the multiple extract tool from this GSD workshop .Once extracted we join all the surfaces by using the tool join which is also in Generative Shape Design workshop.

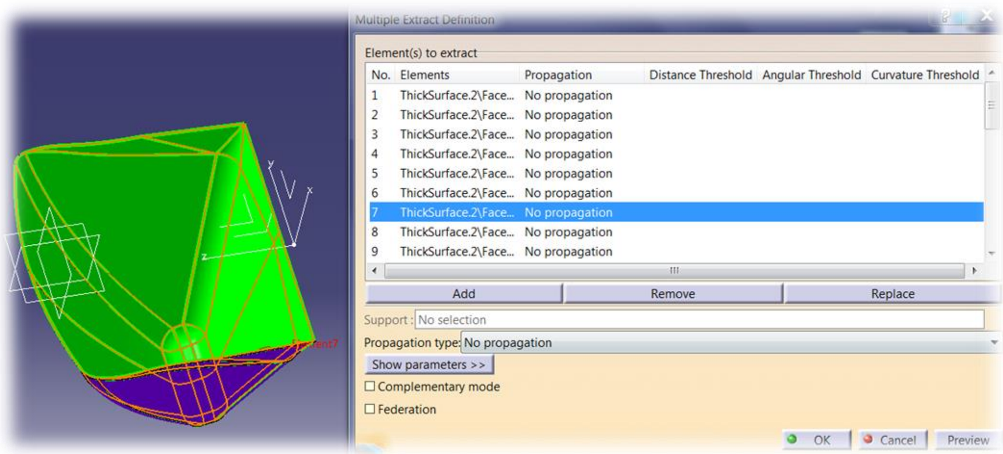


Figure 5. The extraction surface problem

As we observe on the picture, we have to take care about selection of the entire interior surface of the nosecone without selecting the external surfaces.

The second step of the male mould building process was to offset the joined surfaces at 4.8mm (three times the thickness of the mould) distance inside and from the extracted surface to generate space required in the task book.

The challenge was to find out the right value to offset the joined surface without damaging our extracted Nosecone .This proceeded by trial and error. Different values have been tested until finding the value which is a compromise between nosecone dimension and the 4.8 mm gap (three times the nosecone thickness) demanded in the task book.

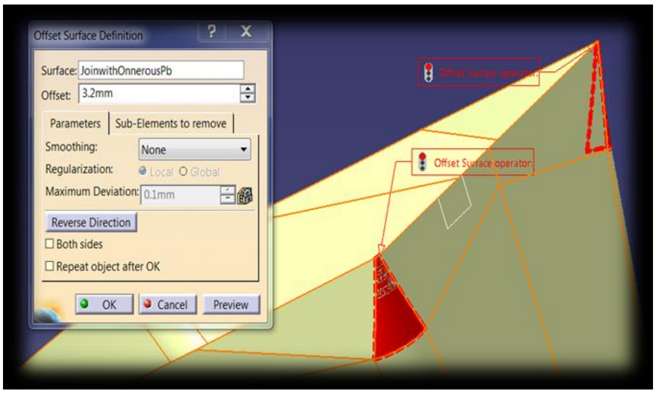


Figure 5.1-The 3.2 mm offset trial value

The 3.2 mm value of the offset nosecone surface show us erroneous zones

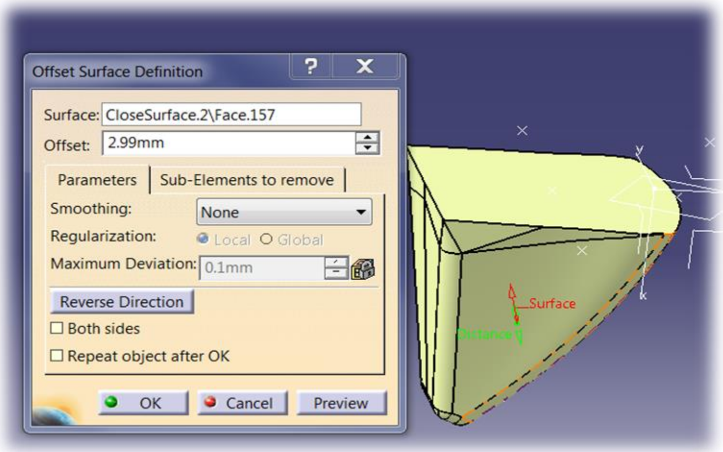


Figure 5.2-. The 2.99mm offset trial value

Testing the 2.99 mm value of the offset nosecone surface resulted in erroneous zones (in red, see figure 5.2

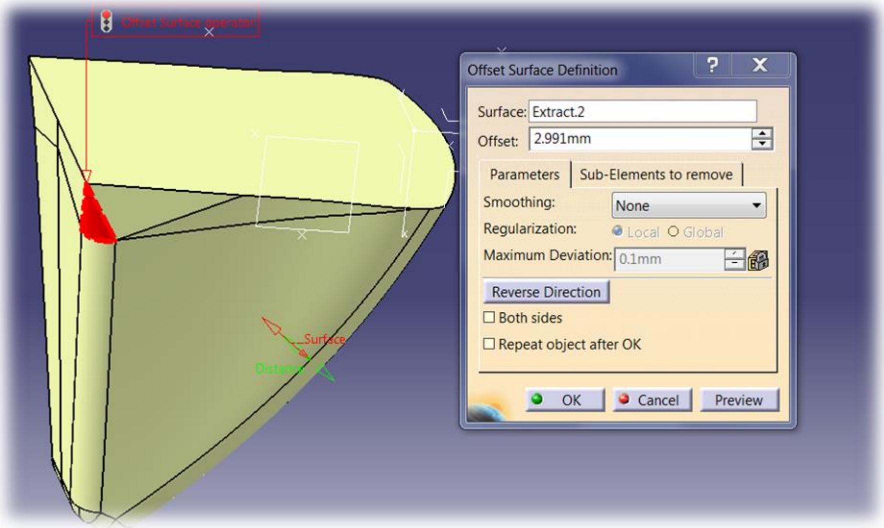


Figure-5.3. 2.991 erroneous value

According to results above, 2.991 mm is an erroneous value

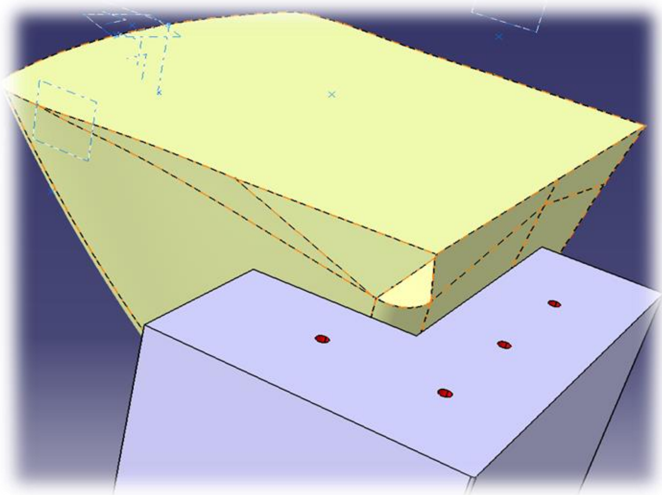


Figure-5.5. -The 4.8mm offset destroyed certain surfaces of the mould which were also erroneous zones

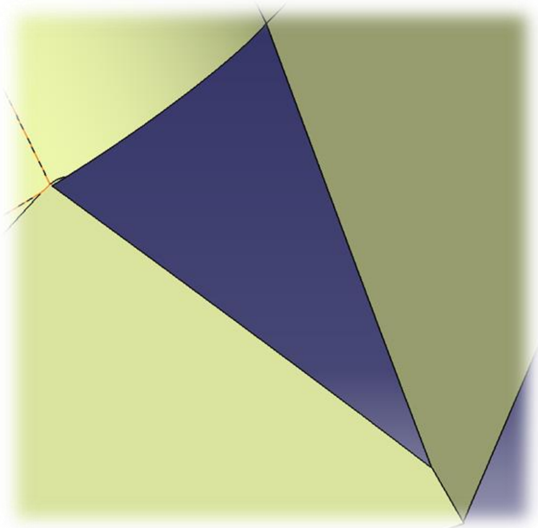


Figure 6 –Nosecone absent surface

After the offsetting operations for the values 3.2 mm, 2.99mm, 4.8mm, some parts of the nosecone model had been removed. As a result, Nosecone dimensions were erroneous.

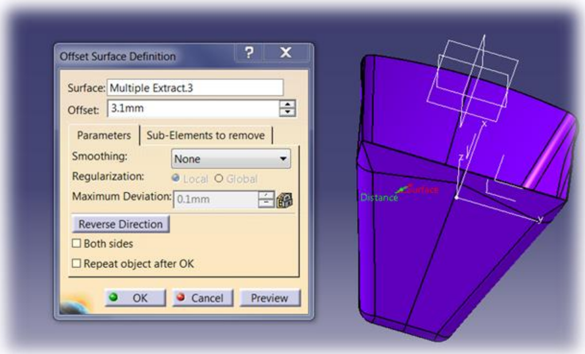


Figure-7.the 3.1mm offset trial value

After the offset process (see Figure-7) the offset from trial value 3.1mm did not generate any errors. When offsetting with this value on an extracted joined surface, there is no damage.

Thus, 3.1mm value is the compromised value that we were looking for. Therefore the only solution to respect the task book is to rebuild the erroneous surface.

Via the fill tool of the Generative Shape Design we selected the joined surface upper contour and filled in the whole and closed nosecone surface extracted. As we wanted to close the surface together which are the filled one and the joined one, we used one more time the join tool of GSD workshop. Then, we successfully joined together, the former joined surface and the filled one.(see figures 7.1, 7.20 below).

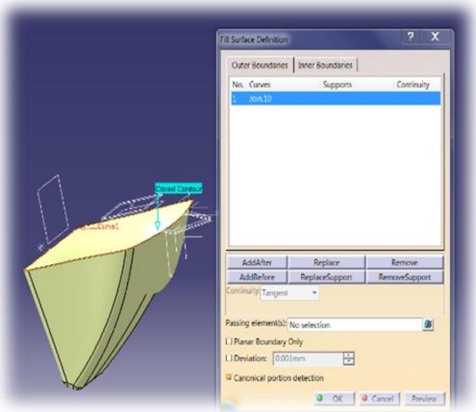


Figure-7.1. filling surface problem

So far from the joining process. At the stage we reflected the main issue affecting the outcome of the joining process was not being identified by the software CATIA V5

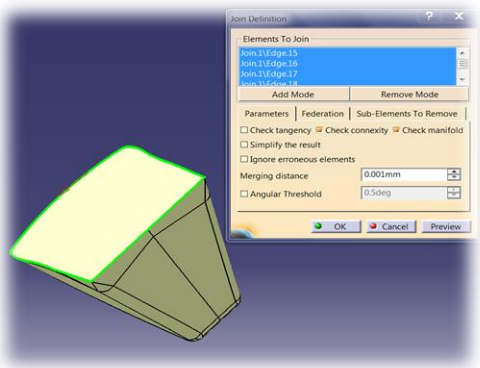


Figure-7.2.Closing contour extracted

On this picture we can observe that the whole entire superior part has been retained with the obligation to use the 3.1mm offset value.

This picture shows us what happened when we filled the nosecone contour without carefully extracted this contour. We can observe that the nosecone shape has been deformed. After extracting and joining contours, it is a lot easier to close the upper and lower surfaces.

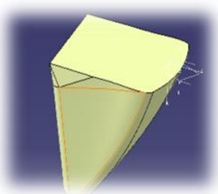


Figure 7.3- Surface re-constructed trial

In figure 7.3, we can observe a cut part of the nosecone has been rebuilt by using construction lines, and the filling tool of CATIA V5.

The initial shape has been deeply modified. Indeed, curves formed from triangular shapes were absent (see figure 6).

Our Challenge was to rebuild a corner surface.

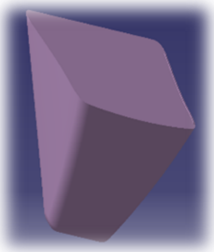


Figure 8-Example of a male mould with volume generated

- This volume has been generated by the Tool close surface from The Generative the Shape Design workshop of CATIV 5. On this picture the entire erroneous surfaces have been removed.

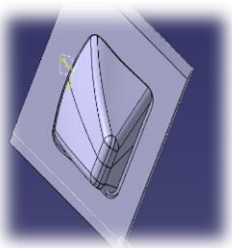


Figure -8.1. The male mould

- Once we created the male mould by filling shape.
  We have to adapt. The sizes of lid which also can be use as lid to the female one.

To use the tool Close surface from Part Design workshop of CATIA V5.We also can generate can use this tool from GSD to generate a volume.

Regarding our offset testing that we have enumerated, the only solution we have found out to generate the 4.8mm space between female mould and the male mould was to rebuild the erroneous surface of nosecone.

Female Mould: The Cavity Design

On this part of our work, we are going to take care about how to create a cavity which is actuality the female mould. We use the joined surface from the nosecone. Then we insert a new body in de tree. We extrude the sketch in form of cube which will envelop the nosecone shape.
By using the tool split from GSD we could split nosecone shape from the second element in shape of parallelogram. At the end of the operation we obtained a cavity.

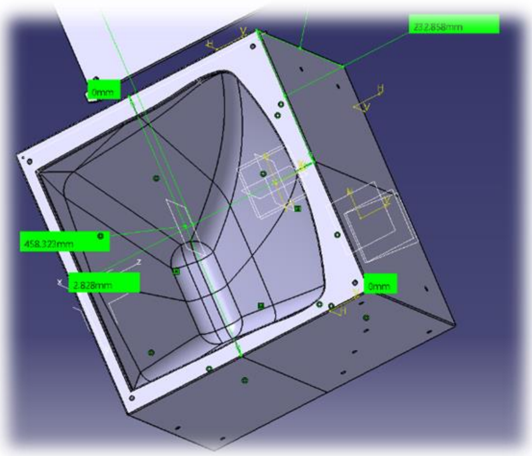


Figure-8.3- The female mould 2014: The Nosecone 2014 Cavity.

The problem with this female is that we have to imagine in which circumstance the nosecone has to be put in the oven. We observe that on the figure 13 that the nosecone cavity is too closed from the edges and there is not enough of space for the oven.

Due the weak thickness between the female cavity the cube .The female mould will crack because of the heat, we also will obtain deformations due the conduction phenomenon.

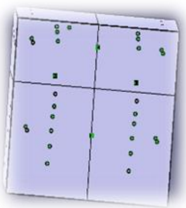


Figure-8.4.Nosecone mould assembled

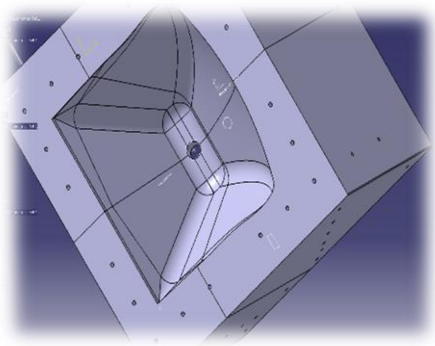


Figure-9-female mould with hole for the resin exes

On the Figure 9.we can observe that the dimensions has been changed.
And holes for bolts are less closed from the female nosecone shape to prevent cracking and deformations due to the heat.

Female mould Nosecone 2014 Dimensions

- These are Nosecone mould 2014 Dimensions
- The length mould is: 609.716 mm.
- The widths mould is: 570mm
- The height mould is: 490 mm

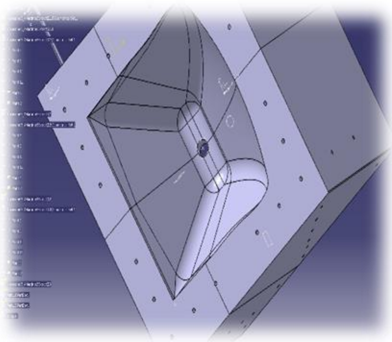


Figure 9.1-Female mould with a hole inside planned for the resin exes.

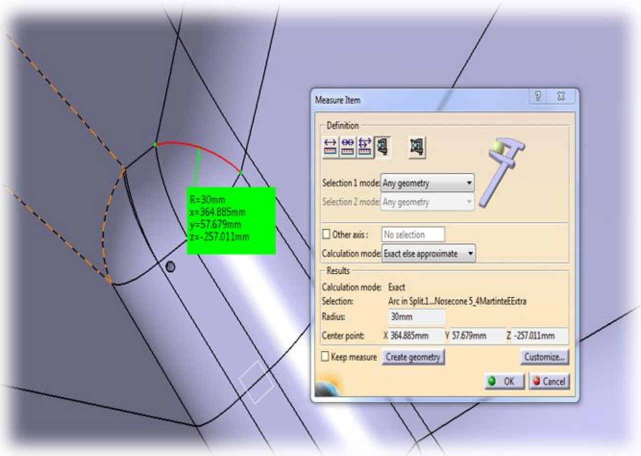


Figure 9.2 R=30mm radius mould edge

Figure-9.2 shows us one of to the female mould radius.

The mould from a manufacturing point of view.

According to a manufacturing point of view, our mould dimension cannot be manufactured because of the size of our mould which are too much according to the milling tool.

Indeed the maximum radius that we got for machining is a 12 mm Diameter tool .that we have for manufacturing is the XL Ball nose and mills (4-fluted, The XL Ball nose end mills from the GUHRING BOOK, code number 12000).

Or one our shortest mould radius was: 30 mm

Moreover we would like to avoid re-entrance due to the size of the mould.

Indeed, the maximum length that we have as milling cutter 150mm. or the height of our mould is 490mm and buying another tool would contribute to exceed the project budget plan.

To solve the problem we could cut the female mould on 8 blocks which will enhance the risk of cantering problems or to scale the mould and cost.

After seeing the most recent nosecone sizes we decided to pay more attention on it to reduce the cost of manufacturing. 2015 Nosecone dimensions were shorter that those 2014. Therefore his mould would be shorter and less expensive build as well. Let’s develop at the next the CAD 2015 mould model.

CAD 2015 Model Investigation

Our choice: **Nosecone 2015 and Nosecone 2016**

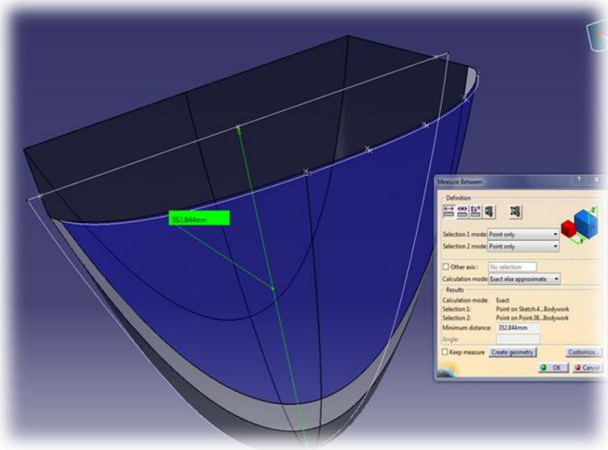


Figure-10 -Nosecone designed By Motorsport student from the Promotion 2015-

Those are Nosecone 2015 dimensions:

- ✓ The width: 420 mm
- ✓ The up middle width: 481.769mm
- ✓ Up width: 395.358mm
- ✓ The high: 352.844mm

Those are the 2015 Nosecone female mould Dimensions.

- ✓ The width: 570mm
- ✓ The length: 680 mm
- ✓ High: 401 mm

Design Methodology

To Design male mould 2015, the methodology is exactly the same than to design the male mould nosecone 2014.The difference were on the mould reconstruction step once the nosecone is damaged by offsetting step.

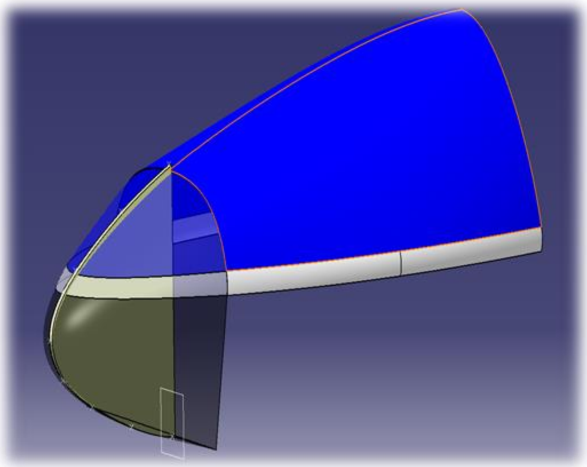


Figure 12- Nosecone 2015 hidden part revealed

To know the thickness of the Nosecone 2015, we got to go at the hidden part of CATIA V5.Thus we could determine that the nosecone 2015 thickness is 1.2mm by revealing the upper, hidden skin of the nosecone.

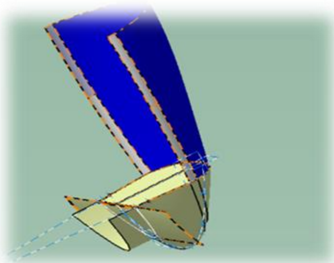


Figure 13-The offseted Nosecone and its construction plans

We just rebuilt the differents the surface which were damaged by using constructions operations which are intersection of lines extrusions. We could the intersections of generated surface to rebuild our offseted surface of 3.6mm (See figure-13, below)

Nosecone 2015 mould Sizing

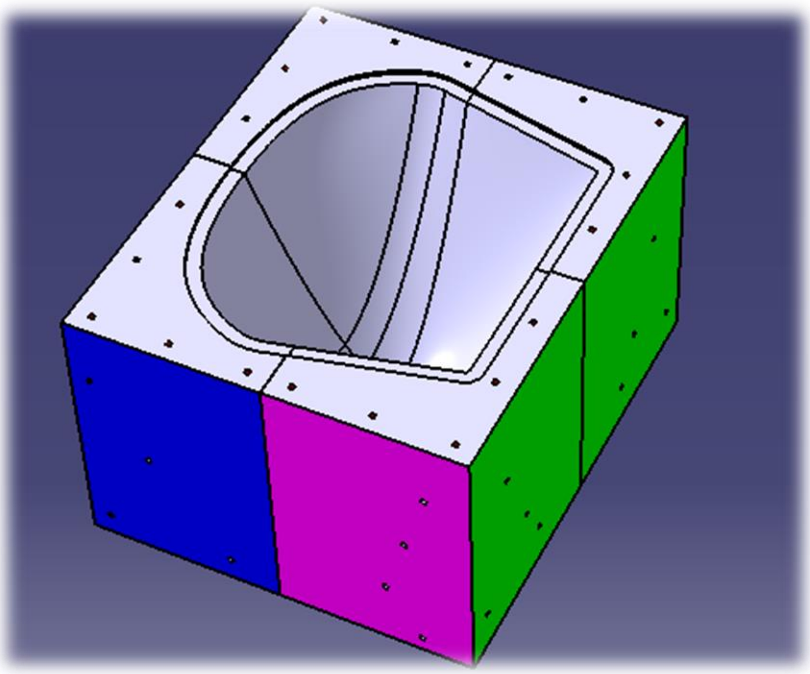


Figure14-Female shape

Mould components

4 Blocks composing the female mould named**:**

- Alpha1prim
- Beta2
- Beta1prim
- Alpha2

- 1 male mould see Nosecone dimensions
- 1 Bolt for male mould lid/hat length bolt: 82mm
- 1 bar for the female mould Length: 740mm
- 1 bar for female mould width length bar: 630 mm

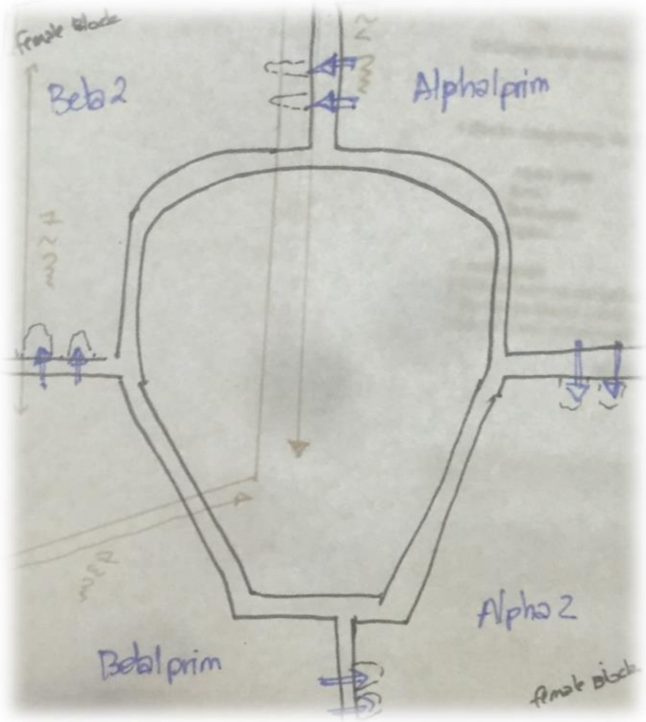


Figure 15- Female components positioning

- **Assembly Design**

Assembly Design framework allows us to check if everything is alright from cad point of view. The designer has to think about the manufacturing and to be aware that the CAD word is a theoretical environment.

During the machining of the object, milling tools, machines, and codes program allows mistake, and mistakes are real because of machining due to the aggregates and scraping.

During the manufacturing process. The material will vibrate during the manufacturing process, the block of aluminium cylinder will be submit to a frequency. Therefore the percentage of mistake cannot be closed of 0%.

Once each part of part of the design mould has been assembled, the main and idea was to think the bars would assure the stability of our mould. This is a wrong idea. Bars cannot assure the centering of our mould. That's why dowels with has been added to favour a better stability to our mould Design.

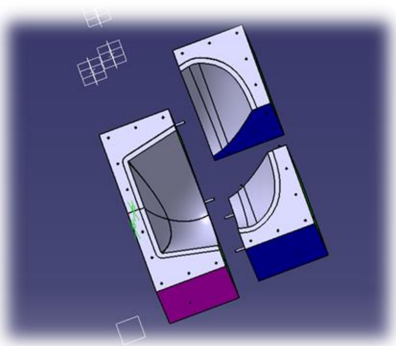


Figure 16-Assembly of Nosecone 2015 block of female mould

For our Dowels we decided to apply with tolerancing h11/d11 which is adapted for the mechanical objects, badly ligned up, with an important freedom movement.
Four our bolt for lids that I have chosen the H7/g6 Tolerancing.
To centre our Nosecone with the female mould we can also add a tolerance of perpandicularity between the surface of the female mould and the surface of the male mould.

Tolerancing Calculations.

Basing on the industrial mechanical engineering book.

- Let's consider the bolts diameter before the one fifth scaling:

For our Bolts:

6H7=6+0.015=6.015
6g6=6-0.09=6.91

6.91mm is the diameter adapted for the bolts. We consider the diameter of the holes on lids is 6mm.
Di=Diameter

The following calculus is going to allow us to no apply tolerancing on bolt for lid but only and directly on holes for dowel.
Di6H7g6=Di6 (15,-9) =Di6 (24, 0) which is the bolt tolerancing for lid
Concerning our dowels:

- DiH11=130-40=90
- 5H11=5+0.090=5.090.
- 5d11=5-0.090=4.91
- 5H11d11=5+0.018=5.18

Therefore 5.18mm is the diameter adapted for the dowel's holes if the dowel's diameter is: 5mm.

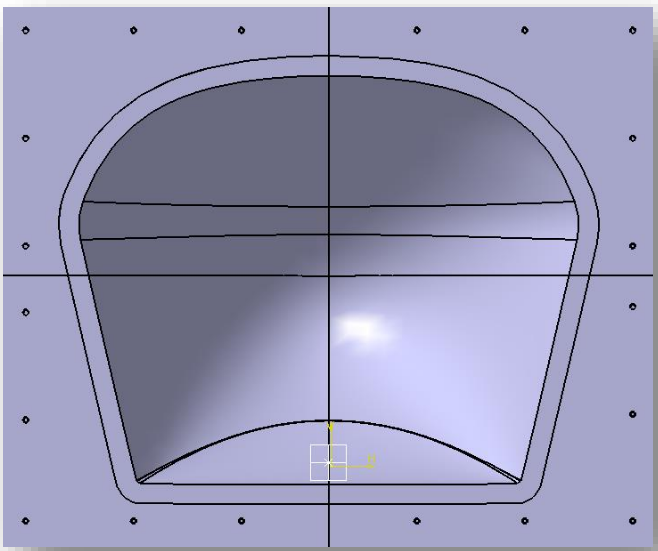


Figure 16-1.1-Female shape 1

Retain container Design

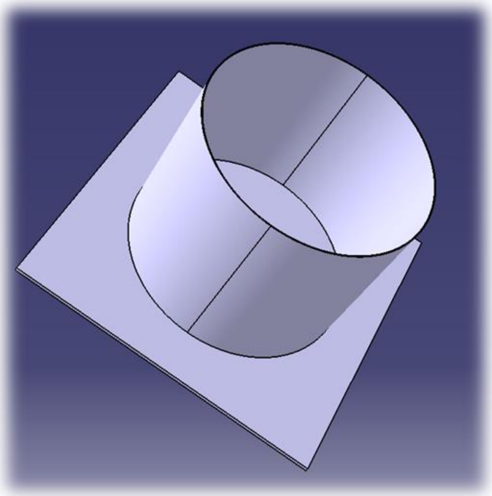


Figure 16.2-Retain container 1 for the bladder moulding

Figure 16.3-Retain container2

In order to experiment the Bladder techniques to get the most adatable aerodynamic shape for Nosecone. Two retain containers has been designed. The retain container 2 is a disposable container and quickly built up. The retain container 1 is on aluminium and durable.Its manufacturing is longer.

These questions have been addressed to the manufacturers and their answers are presented below.

- How much the material is going to cost (per kg)? We are awaiting quotation based on drawing **s**izes. Prices can fluctuate.

- How much is going to cost to be manufactured?
  Not known yet. How long it is going to take? Not known yet.

About the retain container, how much it is going to cost to buy the material for? The retainer will cost approximately £75 +VAT 20% in materials.

- How much it is going to take to be manufactured? The retainer could be made in less than one day. One day costs approximately £450+VAT in technical assistance/labour/equipment time.

- What is the cost estimation n about each material? (Male mould, female mould... bolt, bars....) Each material? cost less than £10

The more material, i.e., the larger the mould, the greater the costs.
The supplier sells material in standard sizes, for example 50mm x50mm, 75mm x 75mm, 100mm x 75mm, etc. etc.

We have to purchase the material *size above* your maximum product sizes, plus any work holding/clamping requirements.

So, if we wanted to machine a component measuring 125mm x 100mm, we would need to purchase material the next available size up, measuring maybe 150mm x 125mm.
The specification of the material will also have an effect on price.

Harrier Jet Jump Flap Investigation

Project Objectives

We would like to develop the most solid composite Harrier Jump flap. We need a CAD model to determine the appropriate dimensions. Indeed the Harrier Jump Flap is a 50 years old one and many parameters such as light, paintings destort in our results. Thus, our task is to pick up a maximum amount of informations from the full size flap in order realize a reliable Airfoil structure composite development

1) <u>Datas recovery</u>

The first step study was to get the maximum of real dimensions directly from the flap by using measuring tools. (Ruler, vernicaliber…. Etc.).We know that whatever the results we got from experiences we will compare that with we the results obtained by hand.
In order to get a Computer Design of the Flap we used metrology tools.

Figure 17-Flap scan's process

Once the full size flap has been scanned, a metrologist cleaned the CAD model by using the software Polyworks and generate a cloud of points that we could import on CATIA V5.

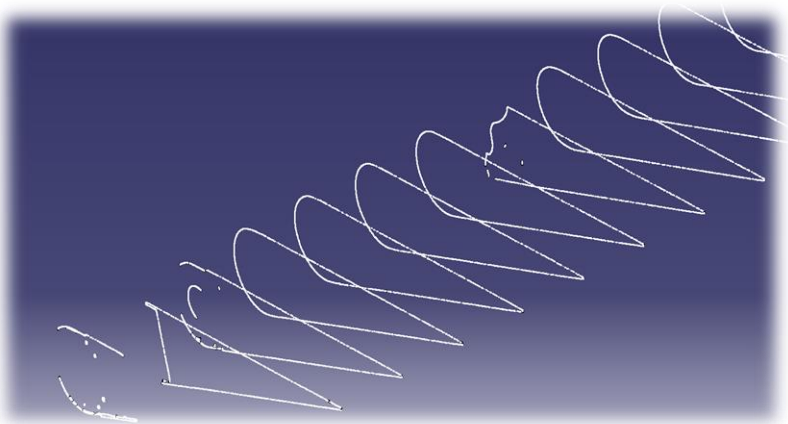


Figure 18- Flap Sections of points imported from Polyworks as an igs file.

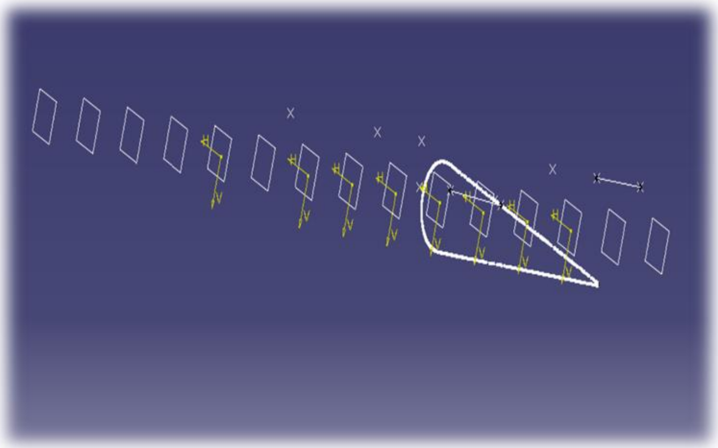


Figure 18.1-Airfoil prototype

We chose one of the most completed file

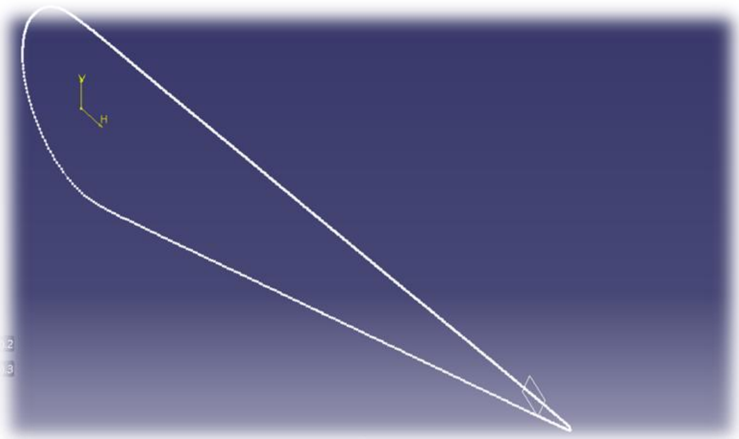


Figure 18.2- An Harrier Jump Airfoil prototype

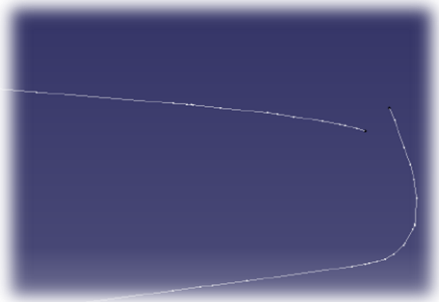


Figure 18.3-Trailing edge damaged by the scan on airfoil

We can observe 15 profiles generated by the scan. The metrologist make the reconstruction realisable by cutting CAD Flap model on 15 sections. Airfoils points could remove to not stop the reconstruction.

Reconstruction flap trials

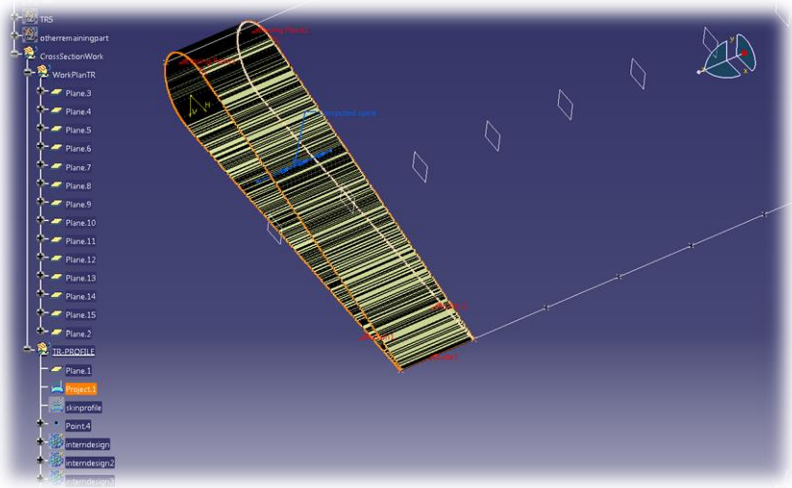


Figure 19 -Multisection surface flap creation

- In order to rebuild the major part of the flap, we decided to choose one of them, that's we use duplicate by using equidistant plan and the tool Multisection ion surface of CATIA V5.

The line direction that we will use is those joining the trailing edges two airfoils.
We know that full size model measure 1500 mm. Basing on the scan the distance between two airfoils is 100mm.Thus we can create plans at 100 mm each other's and make 15 projections on 15 plans
Then we tool Mutltisection surface of GSD to generate surfaces between airfoils a plan.
The problem caused by this method is CATIA V5 converts points on lines and we observe interferences between surfaces. This interference is to the lines.

Reverse engineering

In order to get the maximum dimensions about the Harrier Jet Jump Flap Design, we will work in the Digitized Shape Editor CATIA V5 framework.

To do the reverse engineering once the STL file has been imported, we observe that we cannot do any accurate drawings with lines with operations on this. The is a film of protection forbidding all direct actions with the model.

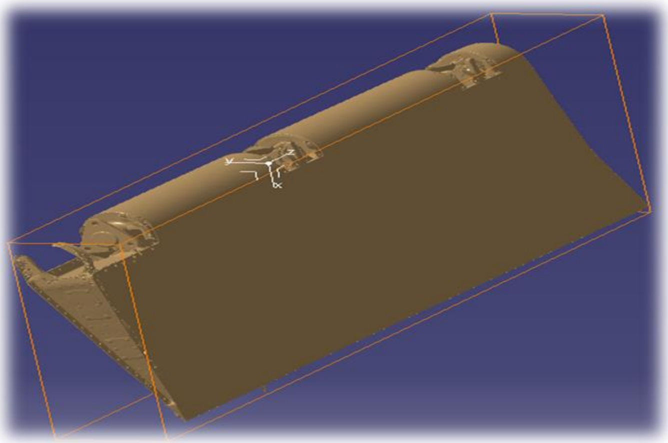


Figure 20 -Window protection flap

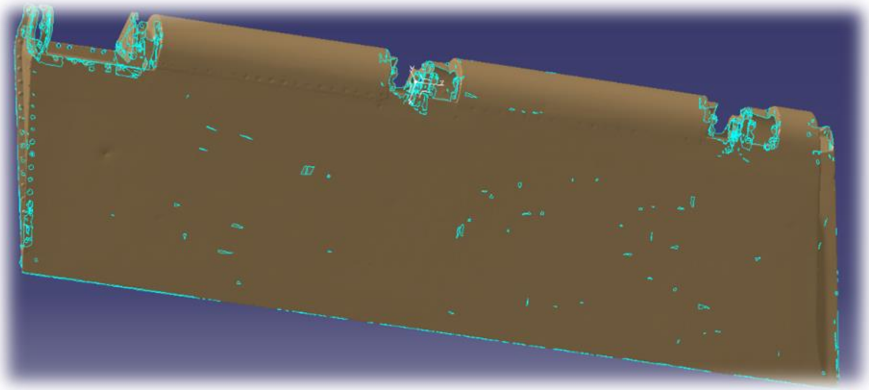


Figure 20.1 -Harrier jump Flap STL- Prototype

To generate the cloud area, we activated the cloud points and the mesh Area.
After testing we realized that weak meshing nodes (number 4 or 5) generate weak clouds of points which is not good if we want to get the maximum information.
The number which was chosen after testing was 25/30.
Contrasting with the finite element Analysis process, weak mesh would be useless because we are doing a reconstruction process not a study of deformation of strength applied on a flap.
Afterwards, we selected to clean the mesh area to remove the erroneous points from the model.
Once, those has been removed we test rebuilded the flap surface by clicking the interface for a Quick Surface reconstruction. We realized that we cannot realize the process because ofnon-connected zones. As a consequence we have to generate lines. Those lines will help use to re-create the Harrier Jump skin surfaces

- <u>Generate a construction lines from all the directions</u>

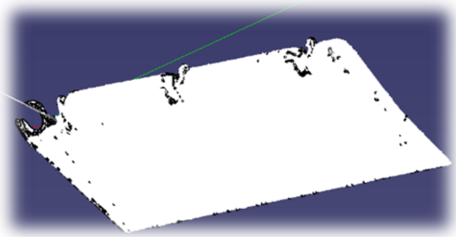


Figure 20.2-multi-directional Flap sections

In order to generate lines from constructions we select XYZ Automatic Planar Section, in digitized shape Editor Framework. This tool will allow us to devellop the flap sections by an optimizing process. (See figure 13)

2) <u>Reconstruction of GSD Model</u>

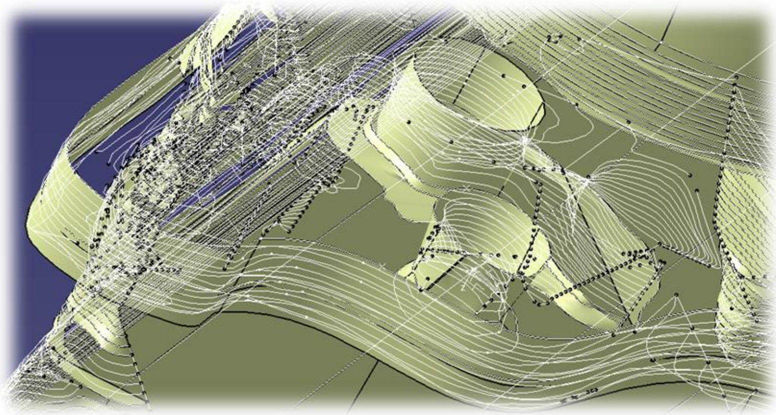


Figure 20.3-Example of Flap complex shape to rebuild

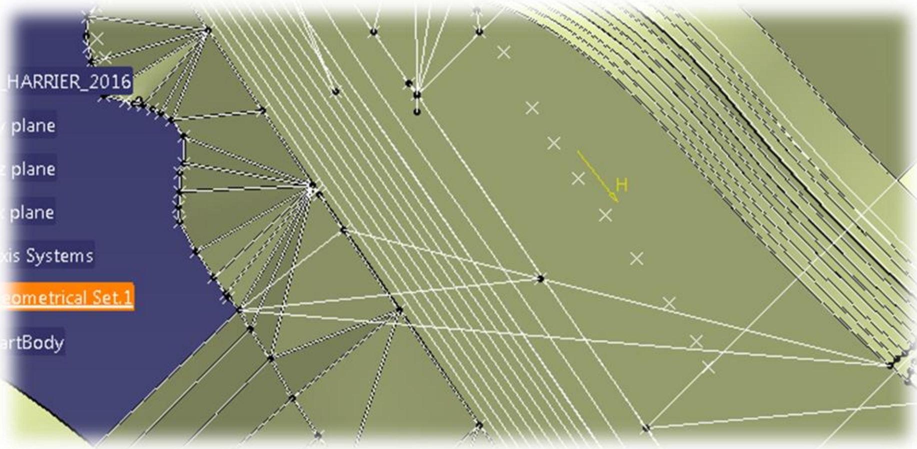


Figure 21- Design of the middle edge flap

How the flaps datas informations have been collected?

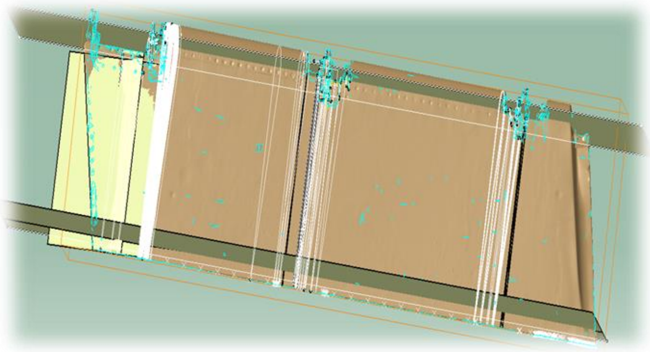


Figure 21.1- Harrier Jump Flap and critical zones

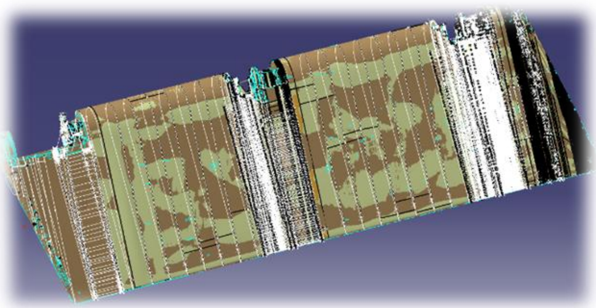


Figure 21.2- Flaps datas collection

To collect a maximum of flap's information, lines and surfaces has been transferred into a same file:The STLversion. Thus, the flap design reveals critical zones. Endeed, the white lines reveals that the flap has been descreted on critical zones in order to gather the maximum of information on this area.

Via the CAD model and the Drafting Part of CATIA, we could generate many drawings (See the Figure which is one of our drawing).

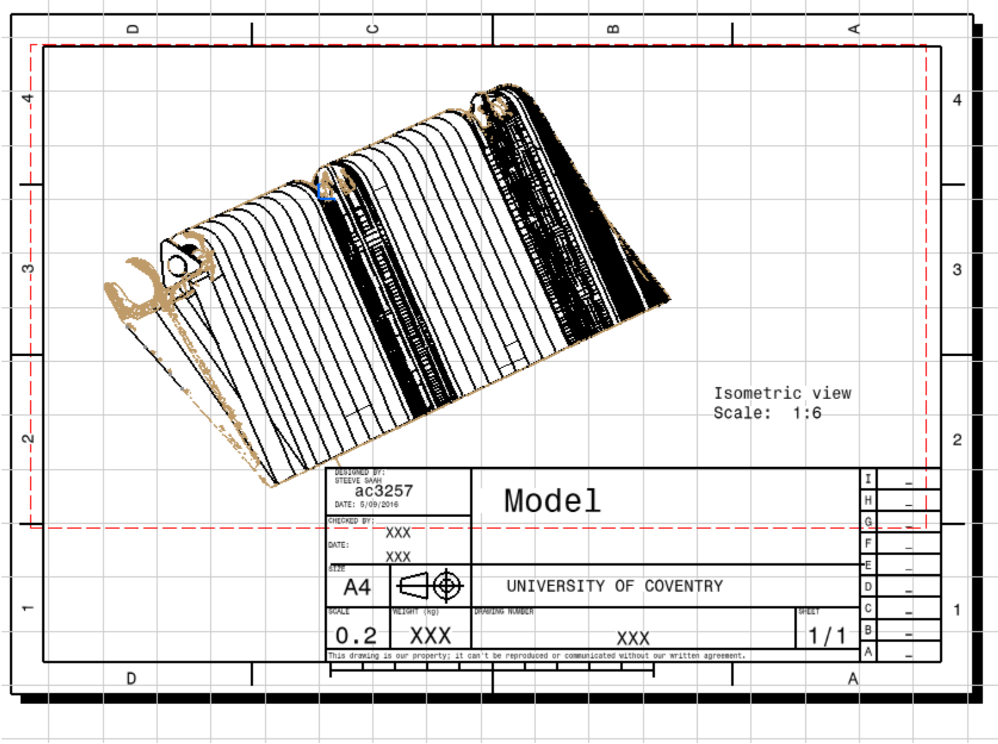


Figure 21.3-Female shape

CONCLUSION

To know the design significance in Bladder moulding techniques, we proceeded by trials an error to conclude that each Nosecone has a specific shape and Design. This, has a repercussion on the material that we would like to devellop on it. This contribute to the composite characteristic.

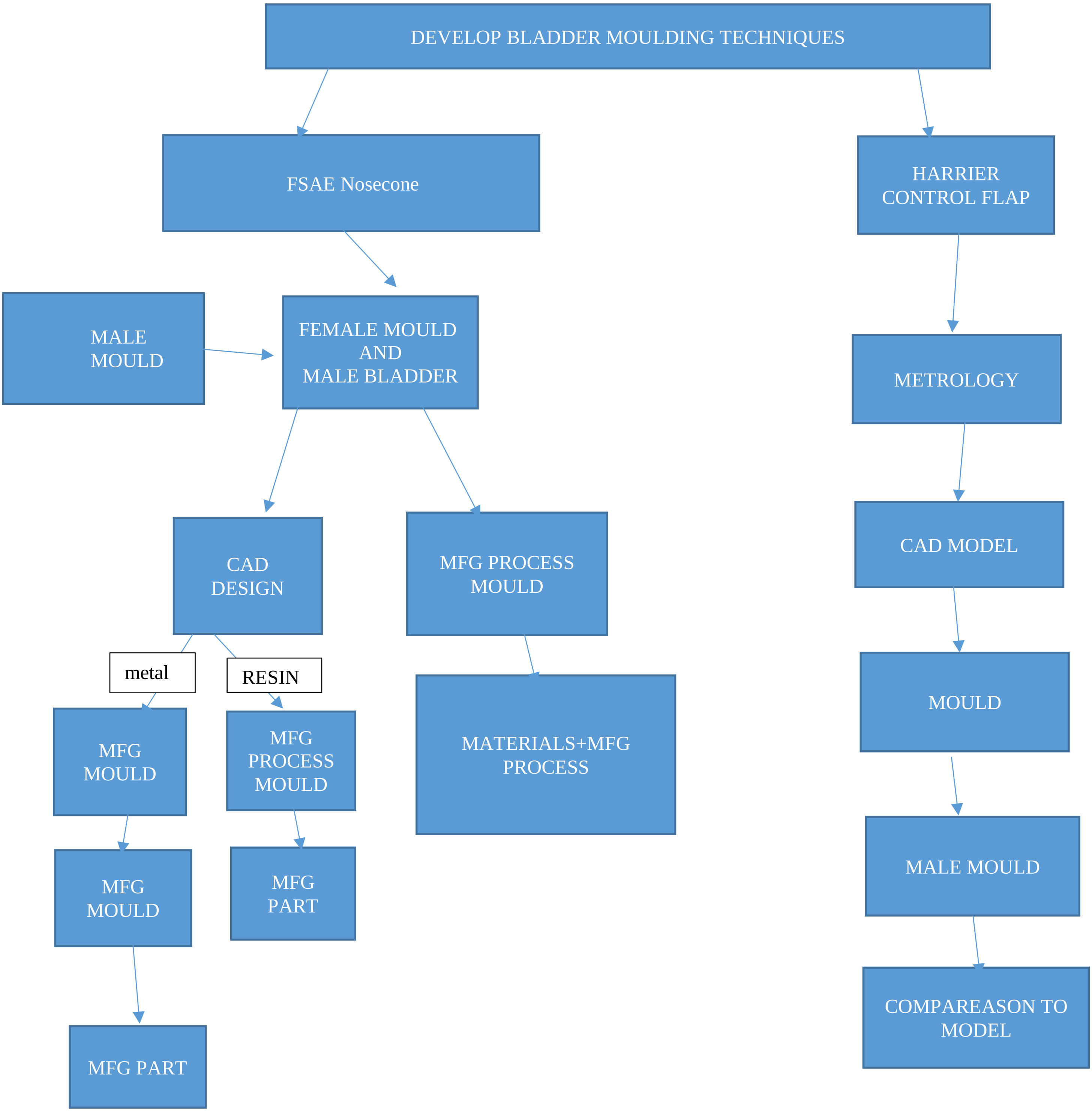
Thus, the nosecone reconstruction operation was necessary to get male mould footprint.

A Disposable female Nosecone is conceivable for the Bladder moulding Process without actually keeping the initial shape.

The Mould designed on this project is a long term product which can be use massively and respect the task book mentioned at beginning of our study: the gap thickness between the female and male mould has to be three three time our Nosecone's thickness.

The main question about our Flap dimension Research, was, how to generate dimensions from our original data which was The Harrier Jump flap scanned model by considering materials damages due to the flap maintenance. This report demonstrates that the Reverse engineering was the most appropriate method and allows us to the develop a Flap composite version by comparing dimensions from the CAD Datas with the real Harrier jump flap dimensions

The Nosecone and the Harrier Jump Composite Development by manufacturing process will be our next Study purpose.



MFG: MANUFACTURING

11/18 tasks of the project has been achieved June to September.
We are going to achieve the other task from September to November.